

Radiation Effects and Defects in Solids

Incorporating Plasma Science and Plasma Technology

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/grad20

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To cite this article: H. Bsharat, Igor Voldiner, B. Ph. van Milligen & C. Xiao (2024) Conditional cross-correlation analysis of floating potential fluctuations in the TJ-II stellarator, Radiation Effects and Defects in Solids, 179:11-12, 1527-1532, DOI: [10.1080/10420150.2024.2434503](https://doi.org/10.1080/10420150.2024.2434503)

To link to this article: <https://doi.org/10.1080/10420150.2024.2434503>



Published online: 31 Dec 2024.



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Conditional cross-correlation analysis of floating potential fluctuations in the TJ-II stellarator

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ABSTRACT

The conditional nonlinear cross-correlation technique has been used to analyze the floating potential fluctuations measured by a radially distributed rake probe array during the electrode biasing experiments on the TJ-II stellarator. Preliminary results suggest that the propagation direction of the turbulence energy changes from outwards in the case without biasing to inwards when biasing is applied.

KEYWORDS

Stellarator; conditional nonlinear cross-correlation; electrode biasing; turbulence

1. Introduction

Linear cross-correlation analyses are often used to reveal the phase propagation direction of fluctuations in fusion plasmas. Conditional cross-correlation analysis of fully developed turbulence measured at two locations can reveal the propagation direction of the turbulence energy (1) or entropy (2). The algorithm has been applied to the potential fluctuations measured at two radial locations in the TJ-II stellarator during the electrode biasing experiment. In this synopsis, introduction to the algorithm of the conditional cross-correlation analysis will be introduced in Section 2. Experimental setup will be described in Section 3. Finally, Section 4 will present the results which indicate the radial propagation direction of turbulence energy changes from outward without biasing to inward when negative electrode biasing is applied.

2. Conditional cross-correlation

Conditional nonlinear cross-correlation analysis technique can be used for studies of fluctuations and turbulences to reveal the relationship between turbulences measured at two locations (1). If we have two discrete signals x_j and y_j sampled at times $t_j = j\Delta t$, where Δt is the time step between data points, a series of M -dimensional vectors X_i^M and Y_i^M can be constructed as

$$X_i^M = \frac{1}{\sqrt{M}}(x_i, x_{i+m}, \dots, x_{i+(M-1)m}),$$
$$Y_i^M = \frac{1}{\sqrt{M}}(y_i, y_{i+m}, \dots, y_{i+(M-1)m}),$$

where m is an integer, indicating that M delayed data points out of x_i and y_i are selected to form an M -dimensional vector. Usually, the m value is chosen based on the auto-correlation time τ based on $m \sim \frac{\tau}{\Delta t}$. The approach makes sure that the samples in the ensemble are random samples for statistical analyses. N samples or vectors (i.e. $i = 1, 2, \dots, N$) for each signal are formed. Each sample, X_i^M or Y_j^M , corresponds to a point on the corresponding M -dimensional phase space. The number M is called embedded dimensions. The analysis evaluates the relative distances, $\epsilon_{xij} = |X_i^M - Y_j^M|$, and $\epsilon_{yij} = |Y_i^M - Y_j^M|$ ($i \neq j$), between different points in their perspective space. The set condition of distance ϵ is a small quantity compared with the size of the attractor ($\epsilon_{\max}$). The cross-correlation between x_i^M and y_i^M can be estimated by setting different ϵ values for the distance in the phase space of X_i^M . The so-called mean conditional dispersion function can be defined by

$$\sigma_{xy}^M(\epsilon) = \sqrt{\frac{\sum_{i \neq j} |Y_i^M - Y_j^M|^2 H(\epsilon - |X_i^M - X_j^M|)}{\sum_{i \neq j} H(\epsilon - |X_i^M - X_j^M|)}}, \quad (1)$$

where $H(z)$ is the Heaviside step function, whose value is zero for negative arguments $z < 0$ and unit for positive arguments $z > 0$. $\sigma_{xy}^M(\epsilon)$ describes the average sizes of the y -attractor under the condition when the x -attractor size is smaller than a set value, $|X_i^M - X_j^M| < \epsilon$. The correlation coefficient between x and y is expressed as:

$$g_{xy} = \lim_{M \rightarrow \infty} \left(\frac{\epsilon_{90}}{\epsilon_{\max}} \right), \quad (2)$$

where ϵ_{90} is determined from the condition $\sigma(\epsilon_{90}) = 0.9\sigma_{\max}$, and $\epsilon_{\max}$ is the maximum size of the attractor.

$$\epsilon_{\max} = \max(|X_i^M - X_j^M|). \quad (3)$$

The approach to find ϵ_{90} and $\epsilon_{\max}$ will be elaborated in Section 4. To find the propagation direction, we define a directional correlation coefficient S_{xy} :

$$S_{xy} = \frac{g_{xy} - g_{yx}}{g_{yx} + g_{xy}}. \quad (4)$$

If $S_{xy} > 0$, the direction of the energy propagation is from x to y . If $S_{xy} < 0$, the direction is from y to x . The ability of this technique to predict energy transfer (propagation) directions was validated by applying this technique to the numerical solutions to two coupled Van der Pol equations (1).

3. Experiment setup

TJ-II is a flexible stellarator device operated in Ciemat, Madrid (Spain) at Spain's National Fusion Laboratory (3). It has 32 toroidal coils, 4 poloidal coils (2 above and 2 below), and 2 helical coils around the central conductor as shown in the center of Figure 1. The central conductor is inside the toroidal coils. The plasma and vacuum vessel form a helix around it. The combination of these magnetic fields generates bean-shaped magnetic surfaces and its orientation changes with the toroidal angle as shown on the top-left and bottom-right corners in Figure 1.

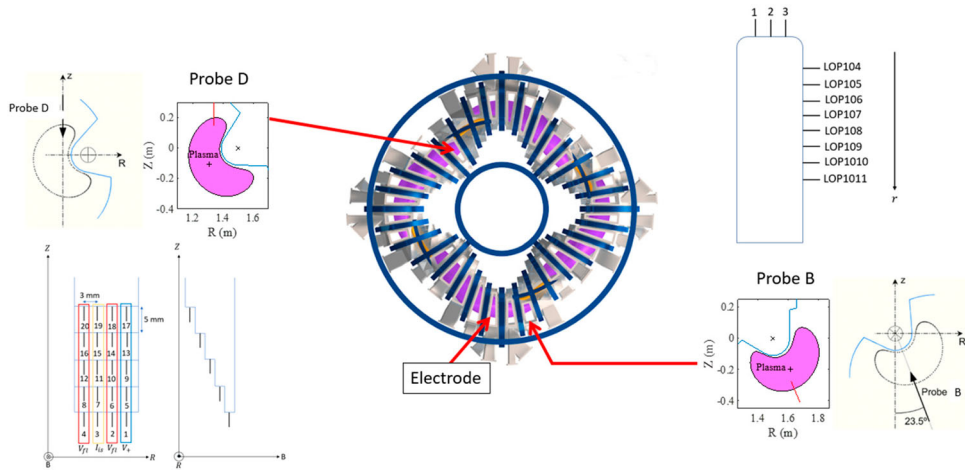


Figure 1. The staircase-like probe D, rake probe B, and the biasing electrode locations during the electrode biasing experiment on TJ-II. The cross-sectional views of the plasma column at the probe insertion locations and the two probes configurations are shown as well. The red line over the plasma cross-sections indicates the ways how the probes are inserted.

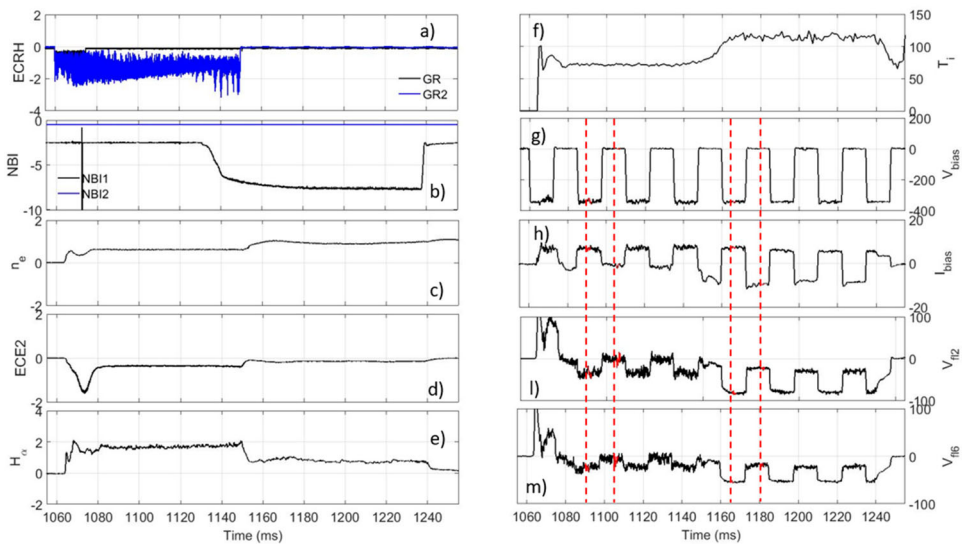


Figure 2. TJ-II discharge waveforms for shot number #43823.

During the electrode biasing experiment, an electrode is inserted into the TJ-II plasma and is biased, with respect to the vacuum vessel, with a square voltage waveform of -350 V in amplitude. The toroidal location of the staircase-like probe D (bottom-left corner in Figure 1), rake probe B (top-right corner in Figure 1) and the biasing electrode during the biasing experiment is also indicated. The purple color shows the plasma in the TJ-II chamber and the red lines are the Langmuir probes position relative to the TJ-II plasma. Probe D enters the plasma vertically from a port on the top. Probe B enters the plasma at an angle from a lower port. The two probes are toroidally separated by approximately 150° .

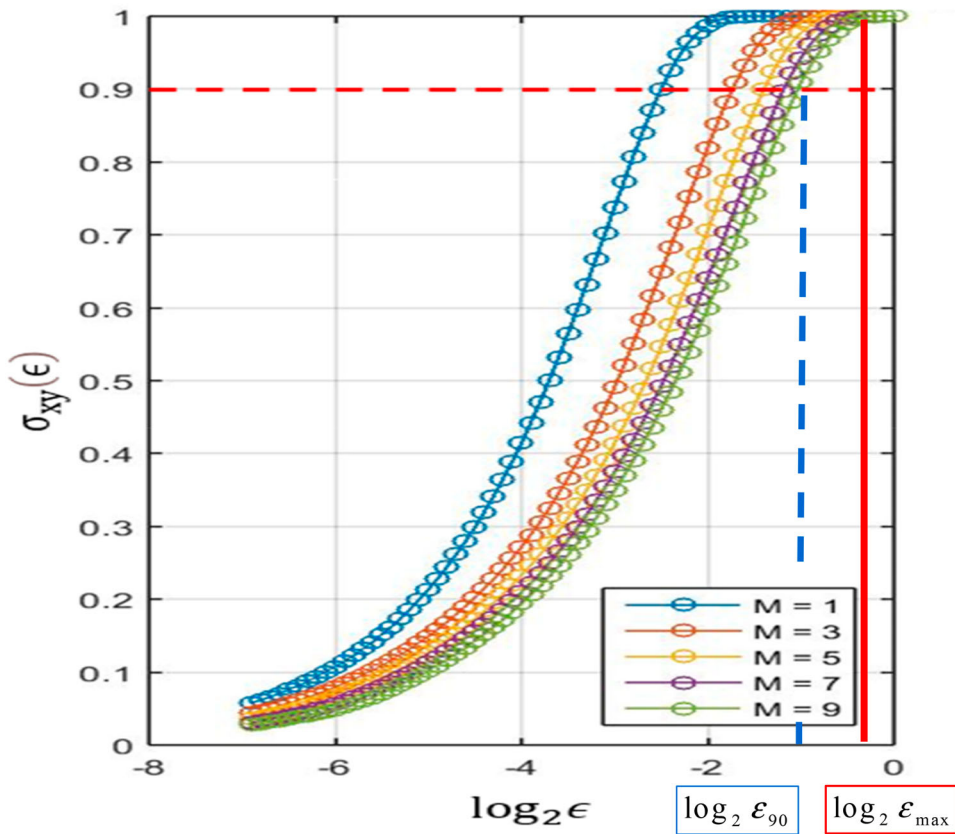


Figure 3. Conditional dispersion curves used to determine the maximum dimension used in nonlinear correlation analysis and to find the ϵ_{90} and $\epsilon_{\max}$ values.

4. Results

Figure 2 shows the waveforms of TJ-II discharge #43823. The discharge is initiated by electron cyclotron resonant heating (ECRH) (panel (a)) and later heated by neutral beam injection (NBI) (panel (b)). Other plasma parameters include (c) normalized density, (e) H_α line radiation intensity, and (f) ion temperature in eV. An H-mode-like discharge phase appears to emerge during NBI, signified by reduction of the H_α line radiation intensity. Rectangular electrode biasing voltage of -350 V and the corresponding biasing current through the plasma are shown in panels (g) and (h), respectively. The floating potentials measured by two rake probe pins are shown in panels (l) and (m).

For the conditional nonlinear correlation analysis, the probe B data was used. The total data length used for this analysis was 4000, and five increasing dimensions $M = 1, 3, 5, 7, 9$ to determine the highest dimensions were chosen. The highest number $M = 9$ is determined after analyzing the dispersion curves of all lower dimensions. Numerically, this number is determined by the convergence of the dispersion curves when the dimension is increased. The results of maximum number of dimensions should be used to find the correlation coefficients g_{xy} and g_{yx} . We choose a semi-log plot to illustrate the dispersion curves

Table 1. Summary of TJ-II conditional nonlinear correlation analyses results at four different times.

Time (ms)	$g_{\tilde{v}_{f3}, \tilde{v}_{f2}}$	$g_{\tilde{v}_{f2}, \tilde{v}_{f3}}$	$S_{\tilde{v}_{f3}, \tilde{v}_{f2}}$	Comments	Propagation direction
1090	0.138	0.043	0.524	No NBIWith biasing	Inward
1105	0.079	0.123	−0.218	No NBINo biasing	Outward
1165	0.203	0.095	0.365	With NBIWith biasing	Inward
1180	0.238	0.260	−0.0442	With NBINo biasing	Outward

$\sigma_{xy}^M(\epsilon)$ against $\log_2 \epsilon$ (the horizontal axis). The dispersion curves can be plotted for different dimensions M . Figure 3 shows a group of dispersion curves $\sigma_{xy}^M(\epsilon)$ for $M = 1, 3, 5, 7, 9$, at a step of 2. When M is large enough, the curves converge to almost the same curve. In the diagram, the blue dashed vertical line shows that we can read the values of $\log_2 \epsilon_{90}$ on the x-axis corresponding to 90% of maximum of dispersion curve for the converging dimension $M = 9$. Similarly, the $\epsilon_{\max}$ value can be read based on the red solid vertical line. We can then calculate the correlation coefficient g_{xy} based on Equation (2). Similarly, we can generate the dispersion curves $\sigma_{yx}^M(\epsilon)$ and determine the correlation coefficient g_{yx} and further determine the directional correlation coefficient S_{xy} .

The conditional correlation analysis between $\tilde{v}_{f2}$ (pin LOP105 in Figure 1) and $\tilde{v}_{f3}$ (pin LOP106) fluctuations of the rake probe B is applied near four different times, indicated by the vertical dashed lines, under different heating and biasing conditions. The conditional cross-correlation coefficients $g_{\tilde{v}_{f2}, \tilde{v}_{f3}}$, $g_{\tilde{v}_{f3}, \tilde{v}_{f2}}$ and the directional correlation coefficient $S_{\tilde{v}_{f3}, \tilde{v}_{f2}}$ are calculated. The results are summarized in Table 1. At the times $t = 1105$ ms and $t = 1180$ ms when the electrode biasing voltage was zero, the signs of $S_{\tilde{v}_{f3}, \tilde{v}_{f2}}$ are negative, suggesting that the turbulence energy propagation direction is outward from pin LOP105 to pin LOP106. In contrast, at $t = 1090$ ms and $t = 1165$ ms with the electrode biasing voltage of -350 V, the signs of $S_{\tilde{v}_{f3}, \tilde{v}_{f2}}$ are positive, indicating that the propagation direction of the turbulence energy is inward from pin LOP106 to pin LOP105.

Acknowledgements

The authors would like to express their gratitude to NSERC, University of Saskatchewan, IAEA CRP F13019, and TJ-II team.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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